

Energy level diagrams of diatomic molecule.

Hydrogen

The wave function is generally taken as a linear combination of atomic orbitals (LCAO) of an electron moving abo. in the field of two nuclei A and B.

$$\Psi = C_A \Psi_A + C_B \Psi_B$$

where Ψ_A and Ψ_B are ^{wave func of} atomic orbitals and C_A and C_B are coefficients chosen to give lowest energy.

In case of homonuclear diatomic molecules such H_2, O_2 equal wts must be given to Ψ_A and Ψ_B on symmetry ground. so that $C_A^2 = C_B^2$ or $C_A = \pm C_B$.

If $C_A = 1$ $C_B = \pm 1$. and hence we have two MO's.

$$\Psi_+ = \Psi_A + \Psi_B \quad \text{and} \quad \Psi_- = \Psi_A - \Psi_B$$

Thus when two atomic orbitals combine together, two MO's are formed one is of lower energy called bonding MO and the other of higher energy is called antibonding MO.

Hydrogen molecule the MO's are ~~the same~~ and the bonding and antibonding MO's are then

$$\Psi_b = H_A(1s) + H_B(1s)$$

$$\Psi_a = H_A(1s) - H_B(1s)$$

(+) sign indicates the ^{increase of} electron density in the region between the nuclei, whereas the (-) sign leads to a decrease of electron density in this region.

Wave function

$$\psi_g = \psi_A(1s) + \psi_B(1s)$$

Atomic orbitals. (2)



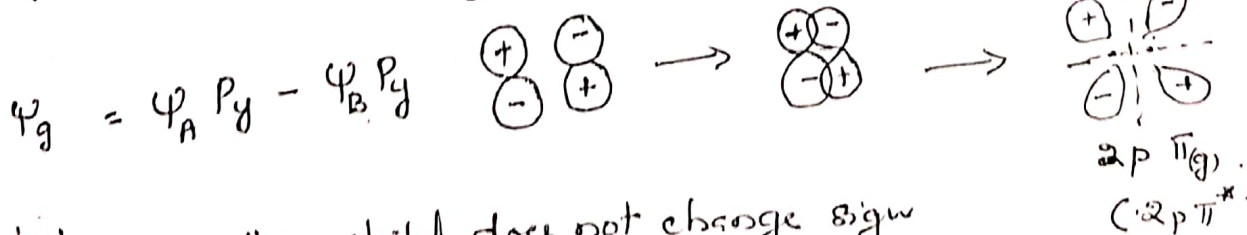
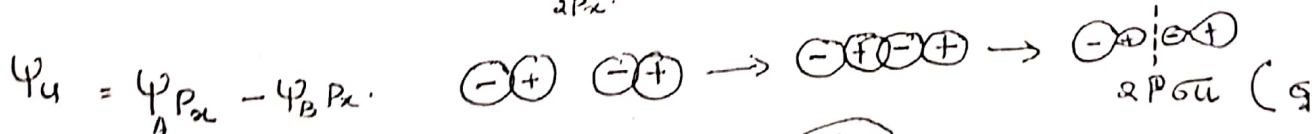
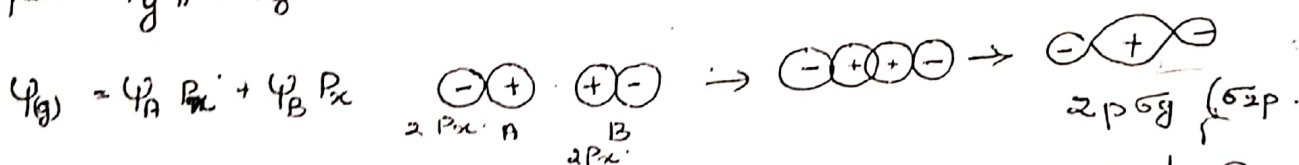
$$\psi_u = \psi_A(1s) - \psi_B(1s)$$



The positive and negative signs in the diagram give the sign of the wave function and the broken lines show the nodal planes. The subscripts g and u are represents the symmetry of the molecular orbital w.r. to inversion centre. 'g' does not change sign on inversion 'u' means it changes sign on inversion operation.

MOs arising from p_x orbitals are called σ orbitals. The two MOs arising from p_y and p_z orbitals are unsymmetrical about the internuclear axis are called π orbitals.

Wave functions which describe more than one orbital are called 'degenerate'. Thus $p_y \pi = p_z \pi$ and the antibonding $p_y^* \pi^* = p_z^* \pi^*$ are doubly degenerate.



'g' means the orbital does not change sign on inversion thro' the centre of symmetry at the midpoint between the two nuclei.

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The molecular orbitals are arranged in the order of their energies as:

$$1s < 1s^* < 2s < 2s^* < 2p_x \sigma < 2p_y \pi = 2p_z \pi < 2p_y \pi^* = 2p_z \pi^* < 2p_x \sigma^*$$

* A similar arrangement exist for 3s to 3p² MO's.

Fundamental ideas of MOT.

1. All the atomic orbitals of interacting atoms are combined in pairs.
2. The atomic orbitals may originally occupied by electrons or not.
3. The combining orbitals must be of same energy.
4. The combining orbitals must have the same symmetry.
5. The number of MO's formed is equal to the number of

Combining AO's.

6. The MO's are arranged in ascending order of energy and all electrons in AOs are redistributed in MO's based on aufbau principle and Pauli's exclusion principle which limits max. number of e's in each MO is two.
7. The bonding orbitals are filled first then antibonding orbitals.

For example in O₂.

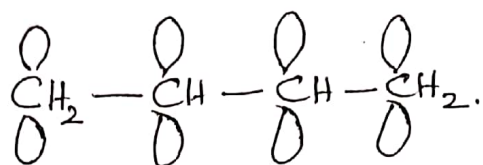
$$8 + 8 = 16 \text{ e}^-$$

$$\sigma_{1s}^2 < \sigma_{1s}^{*2} < \sigma_{2s}^2 < \sigma_{2s}^{*2} < 2p_x \sigma^2 < 2p_y \pi^2 = 2p_z \pi^2 < 2p_y \pi^{*2} = 2p_z \pi^{*2} < 2p_x \sigma^*$$

MOT of Conjugated system. π -molecular orbitals of butadiene.

The HMO.

Consider Butadiene - a conjugated system.



Carbon is in sp^2 hybridisation. Apart from σ frame work, π e's move independently. Sch: eqⁿ can be applied to calculate approximate energy. For this

① Variation method is used. ② Perturbation theory can be used.

In variation method there are two basic approaches to calculate the molecular energies. They are MOT & VBT.

MOT is widely used for treatment of polyatomic molecule in Chemistry

Variation method.

An approximate wave function has to be selected. The selected wave function happens to be the correct wave function only if the minimum value of calculated energy is closer to the ground state energy of the system and we can write for this eqⁿ for calculating the equation energy as.

$$\frac{\int \phi^* \hat{H} \phi d\tau}{\int \phi^* \phi d\tau} \geq E_0$$

$\Psi_{M.O} = c_1 \phi_1 + c_2 \phi_2$ where ϕ_1 & ϕ_2 are (ψ_{1s}) atomic orbital wave fns for electrons in the field of 2 protons

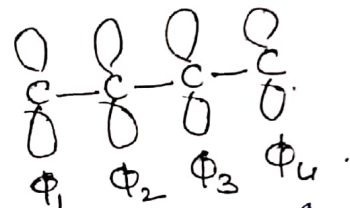
In variation method ⁽⁹⁾ approximate wave function is generally taken as a linear combination of a number of wave functions $\phi_1, \phi_2, \phi_3, \dots$ multiplied by the coeffs. $c_1, c_2, c_3, \dots$ respectively.

$$\psi = c_1 \phi_1 + c_2 \phi_2 + c_3 \phi_3 \dots$$

The coefficients are $c_1, c_2, c_3, \dots$ chosen such that when used in eqn (1) gives minimum value of energy.
method to calculate π m.o energy

HMO of Butadiene.

The trial function is



$$\psi = c_1 \phi_1 + c_2 \phi_2 + c_3 \phi_3 + c_4 \phi_4$$

which when substituted in eqn. $\int \psi^* \hat{H} \psi = E \int \psi^* \psi$ $\hat{H} \psi = E \psi$

$$E = \frac{\int \psi^* \hat{H} \psi d\tau}{\int \psi^* \psi d\tau} \geq E_0$$

and evaluated for energy, different energy states are calculated as.

$$E_4 \text{ ————— } \alpha - 1.618 \beta$$

$$E_3 \text{ ————— } \alpha - 0.618 \beta$$

$$E_2 \text{ ————— } \alpha + 0.618 \beta$$

$$E_1 \text{ ————— } \alpha + 1.618 \beta$$

The resonance energy of butadiene is calculated as $+0.472 \beta$.

continued in last
pg

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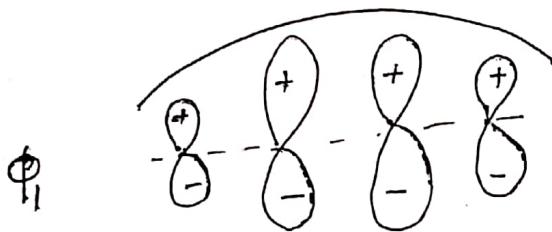
$$\epsilon_4 = \alpha - 1.618\beta$$



$$\epsilon_3 = \alpha - 0.618\beta$$

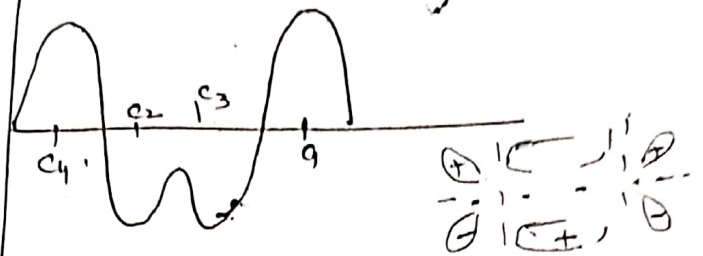
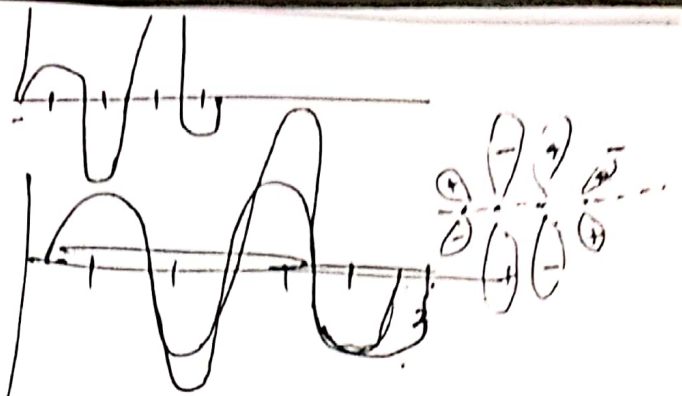


$$\epsilon_2 = \alpha + 0.618\beta$$

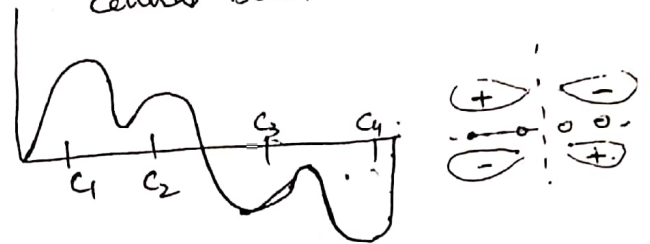


$$\epsilon_1 = \alpha + 1.618\beta$$

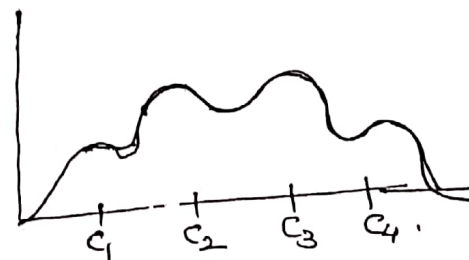
ψ_2
↑



Bonding in outer bonds.
anti bonding in the
central bond.



Bonding in all bonds.
(upper and lower positions)



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$$\psi_1 = 0.372\phi_1 + 0.602\phi_2 + 0.602\phi_3 + 0.372\phi_4.$$

$$\psi_2 = 0.602\phi_1 + 0.372\phi_2 - 0.372\phi_3 - 0.602\phi_4.$$

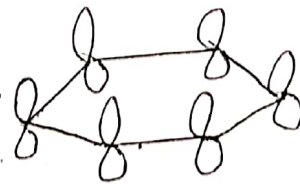
$$\psi_3 = 0.602\phi_1 - 0.372\phi_2 - 0.372\phi_3 + 0.602\phi_4.$$

$$\psi_4 = 0.372\phi_1 - 0.602\phi_2 + 0.602\phi_3 - 0.372\phi_4.$$

wave functions of butadiene.

Pi molecular orbitals of Benzene.

$$\psi = c_1\phi_1 + c_2\phi_2 + c_3\phi_3 + c_4\phi_4 + c_5\phi_5 + c_6\phi_6.$$



Calculated energy values are.

_____	$2 - 2\beta$		
_____	$\alpha + \beta$	$\epsilon_6 = \alpha - 2\beta$	← _____
_____	$\alpha - \beta$	$\epsilon_4, \epsilon_5 = \alpha - \beta, \alpha - \beta$	_____
_____	$\alpha + 2\beta \equiv$	$\epsilon_2, \epsilon_3 = \alpha + \beta, \alpha + \beta$	$\uparrow \downarrow$ _____
_____	$\alpha - \beta$	$\epsilon_1 = \alpha + 2\beta$	$\uparrow \downarrow$ _____
ϵ_1 _____	$\alpha + \beta$		

Resonance energy ^(B) of benzene is called
calculated as 2/3

On solving ^{for} the coefficients $c_1, c_2 \dots c_6$, we get the wave
functions for the six molecular orbitals of benzene as:

$$\psi_1 = \frac{1}{\sqrt{6}} (\phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6)$$

$$\psi_2 = \frac{1}{2} (\phi_2 + \phi_3 - \phi_5 - \phi_6)$$

$$\psi_3 = \frac{1}{\sqrt{12}} (\phi_1 - \phi_2 - 2\phi_3 - \phi_4 + \phi_5 + 2\phi_6)$$

$$\psi_4 = \frac{1}{2} (\phi_1 - \phi_2 + \phi_4 - \phi_6)$$

$$\psi_5 = \frac{1}{\sqrt{12}} (\phi_1 + \phi_2 - 2\phi_3 + \phi_4 + \phi_5 - 2\phi_6)$$

$$\psi_6 = \frac{1}{\sqrt{6}} (\phi_1 - \phi_2 + \phi_3 - \phi_4 + \phi_5 - \phi_6)$$

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Aromaticity (Aromatic character).

out that
MOT points to special electronic stability of the ring compound with 2 ($4 \times 0 + 2$), 6 ($4 \times 1 + 2$), 10 ($4 \times 2 + 2$) etc electrons. This observation was made by Hückel. who gave a rule known as Hückel rule ($4n+2$) rule ($n=0, 1, 2$ etc). This rule states that monocyclic coplanar systems, which contain $(4n+2)$ π electrons possess a large electronic stability due to closed shell configuration. Thus for ~~to~~ systems which contain $(4n+2)$ π electrons are aromatic. where $n=0, 1, 2, 3$ etc.